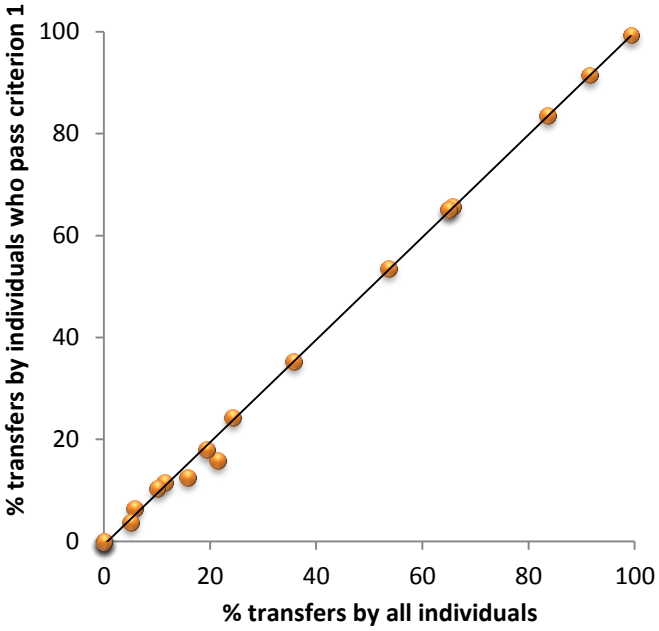


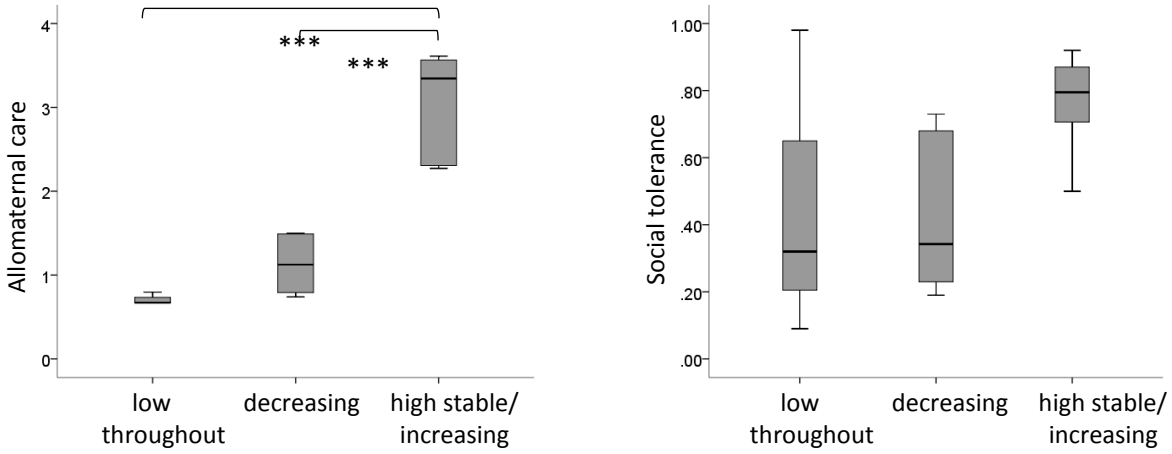
Supplementary Figure 1

Automated version of the group service apparatus used for testing the human children.



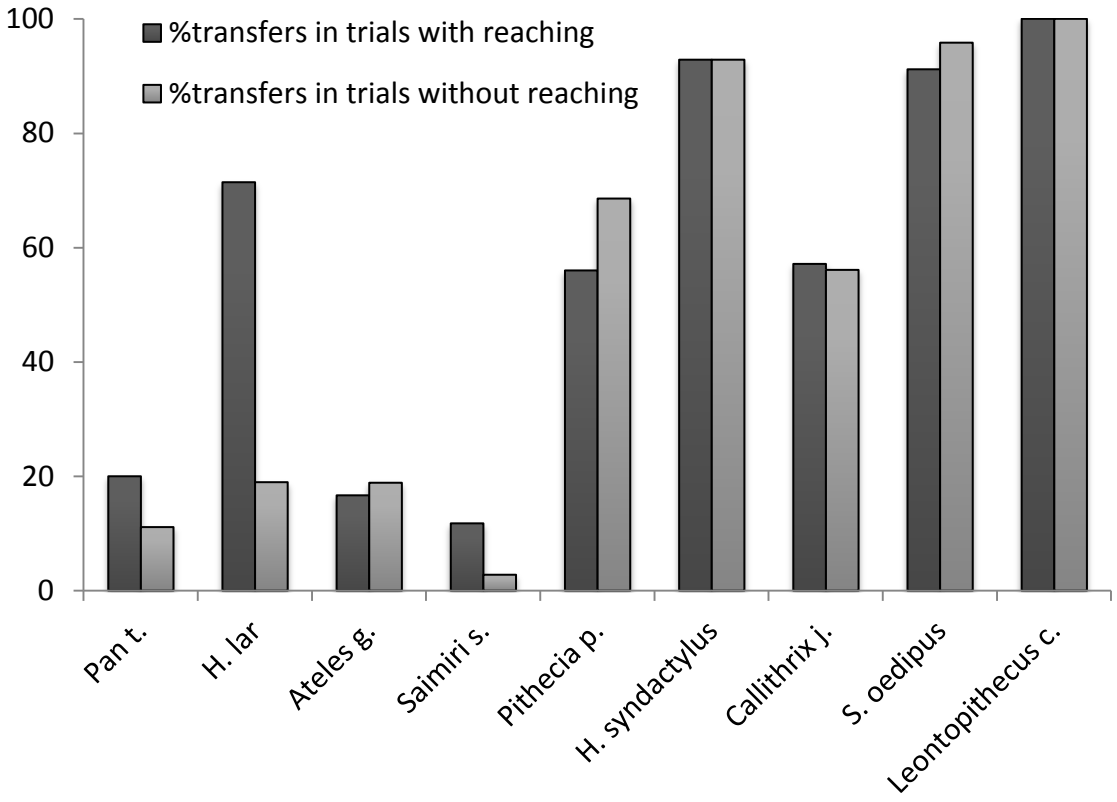
Supplementary Figure 2

Percentage of food transfers in phase IV (test sessions 4 and 5) by all individuals (x-axis) and only by those individuals who passed criterion 1 and pulled significantly more often in test compared to control condition (y-axis). Both measures are strongly correlated; for the comparative analyses, the % of food transfers by individuals who passed crit.1 was used as response variable.

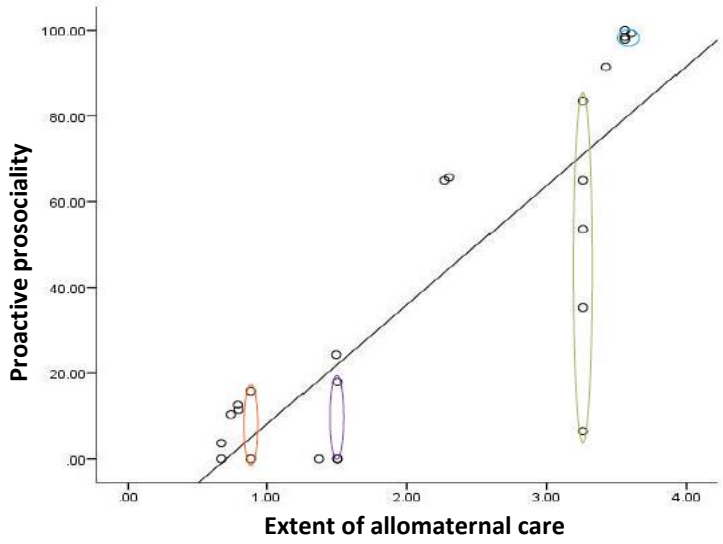


Supplementary Figure 3

Species with sustained deliveries over the five test sessions of Phase IV show significantly higher amounts of allomaternal care, but not social tolerance. Abscissa: progress of deliveries over the five test sessions; low levels throughout (n=3), decreasing levels (n=6) and high stable/increasing (n=6). The box plots represent medians (black horizontal lines), inter-quartile ranges (grey boxes), and minima and maxima (whiskers). ***: $p<0.001$



Supplementary Figure 4: Percentage of transfers in trials with and without reaching (phase IV session 5).



Supplementary Figure 5

Correlation between the extent of allomaternal care and proactive prosociality at the group- rather than species-level. (GLM; response: proactive prosociality, random effect: species. $F_{(13,10)}$=6.24, p=0.003). The group level analyses thus confirm that proactive prosociality is linked to allomaternal care. Orange: spider monkeys (2 groups); purple: capuchin monkeys (3 groups); green: common marmosets (5 groups),blue: human children (3 groups). Only common marmosets show substantial within-species variation in performance. This variation is best explained by group composition, i.e. the amount of allomaternal care available within each group, consistent with the results at the level of between-species variation (results not shown).



Supplementary Figure 6: 5-year-old children in the group service experiment. They start self-organizing without instruction. Picture blurred to ensure anonymity.

Supplementary Table 1: Explanatory variables other than allomaternal care.

Species	Fission-fusion	Cooperative hunting	Male bonds	Female bonds	Pair bonds
Ateles geoffroyi	1	2	1	2	2
Callithrix jacchus	2	2	2	2	1
Cebus apella	2	1	2	1	2
Homo sapiens	1	1	1	2	1
Hylobates lar	2	2	2	2	1
Hylobates syndactylus	2	2	2	2	1
Lemur catta	2	2	2	1	2
Leontopithecus chrysomelas	2	2	2	2	1
Macaca fuscata	2	2	2	1	2
Macaca silenus	2	2	2	1	2
Pan troglodytes	1	1	1	2	2
Pithecia pithecia	2	2	2	2	1
Saguinus oedipus	2	2	2	2	1
Saimiri sciureus	2	2	2	1	2
Varecia variegata	2	2	2	2	1

1 = factor applies to a given species, 2 = factor does not apply.

Note. For *Varecia variegata*, the presence of a fission-fusion social organization has been described. However, their spatial ecology does not adhere to that described for other primate fission-fusion systems but shows distinct features that combines aspects of both female- and bisexually bonded systems of range use[13]. We therefore classified them as not having a fission-fusion organization, but note that classifying them as fission-fusion species does not change the main results.

Supplementary Table 2: Measures of allomaternal care

Species	Groups in Ross & MacLarnon 2000[39]	% allocare in Ross & MacLarnon 2000	Isler & van Schaik 2012[25]	Isler & van Schaik 2012, updated	Broad categories	Sources for update
Ateles geoffroyi	1		0.31	0.88	1	[14,15]
Callithrix jac.	3	46.7	3.88	3.26	3	[16-20]
Cebus apella	2		1.50	1.50	2	
Homo sapiens	3	32.4	-	3.56	3	[21-23]
Hylobates lar	1	0	0.80	0.80	1	
Hylobates synd.	3	50	2.31	2.31	2	
Lemur catta	-		1.42	0.74	1	[24]
Leontopithecus	3	50.8	3.88	3.61	3	[16-20,25]
Macaca_fuscata	1	0	0.67	0.67	1	
Macaca_silenus	1	0	-	0.67	1	[26]
Pan troglodytes	1	3.8	0.79	0.79	1	
Pithecia pithecia	2		1.08	2.27	2	[27-30]
Saguinus oed.	4	68.5	3.43	3.43	3	
Saimiri sciureus	3	30	0.53	1.49	2	[31,32]
Varecia var.	-		1.42	1.37	2	[24]

Note: Within the cooperatively breeding callitrichids, patterns of reproductive failure suggest that *Leontopithecus* and *Saguinus* are more obligate cooperative breeders than *Callithrix* (regarding the necessity of helpers in the group and the importance of experience as helper to successfully reproduce[16-20],) and *Leontopithecus* shows a more specialized form of provisioning by transporting food and putting it down directly in front of the recipient, as well as provisioning of pregnant mothers[25] and of babysitters (data form own colony).

Supplementary Table 3: Correlations between the five measures of allomaternal care.

	Isler & van Schaik 2012, updated	**% allocare from Ross & MacLarnon 2000**	**Broad categories**	**Groups Ross & MacLarnon 2000**
Isler & van Schaik 2012	r=0.907, n=13, p<0.001	r=0.838, n=8, p=0.009	Rho=0.779, n=13, p=0.002	Rho=0.675, n=11, p=0.023
Groups from Ross & MacLarnon 2000	Rho=0.867, n=13, p<0.001	Rho=0.916, n=10, p<0.001	Rho=0.927, n=13, p<0.001	_
Broad categories	Rho=0.941, n=15, p<0.001	r=0.867, n=10, p=0.001	_	_
% allocare from Ross & MacLarnon 2000	r=0.885, n=10, p=0.001	_	_	_

Supplementary Table 4: Phylogenetic least squares (PGLS) models predicting proactive prosociality in nonhuman primates, excluding *Homo sapiens.*

number	**Model selection process:**	K,N	lambda	F	Direction	p-values	AICc	Delta AICc
1	**ln Brain size**	2,14	0	5.08	-	**0.044**	138.3	19.9
2	Cooperative hunting	2,14	0.377	1.53	+	0.240	142.0	23.6
3	Male bonds/Fission fusion	2,14	0.310	1.28	+	0.280	142.2	23.8
4	Female bonds	2,14	0	4.42	-	0.057	138.9	20.5
5	**Pair bonds**	2,14	0	10.54	+	**0.007**	134.4	16
6	**Social tolerance**	2,14	0	7.06	+	**0.021**	136.8	18.4
7	**Allomaternal care**	2,14	0	58.7	+	**<0.001**	118.4	0
8	ln Group size	2,14	0.329	2.95	-	0.111	140.6	22.2
9	ln (Number of immatures +1)	2,14	0	4.77	-	0.05	138.6	20.2
10	**Allomaternal care** Social tolerance Allomaternal care*Social tolerance	4,14	0	19.01	+ + +	**0.011** 0.284 0.341	124.0	5.6
11	**Allomaternal care** Social tolerance	3,14	0	28.02	+ +	**0.0002** 0.551	121.2	2.8
12	**Allomaternal care** Pair bonds Allomaternal care*Pair bonds	4,14	0	18.57	+ + +	**0.047** 0.338 0.362	124.2	5.8
13	**Allomaternal care** Pair bonds	3,14	0	27.63	+ +	**<0.001** 0.630	121.4	3
14	Pair bonds Social tolerance Pair bonds*Social tolerance	4,14	0	3.12	+ + +	0.172 0.697 0.816	141.4	23
15	Pair bonds Social tolerance	3,14	0	5.08	+ +	0.155 0.614	137.4	19
16	**Allomaternal care** ln Brain size Allomaternal care*ln Brain size	4,14	0	20.84	+ + +	**0.002** 0.178 0.567	122.9	4.5
17	**Allomaternal care** ln Brain size	3,14	0	33.04	+ +	**<0.001** 0.177	119.3	0.9
18	**Pair bonds** ln Brain size Pair bonds*ln Brain size	4,14	0	4.29	+ - -	**0.038** 0.287 0.474	139.0	20.6
19	**Pair bonds** Ln Brain size	3,14	0.170	6.68	+ -	**0.034** 0.221	135.7	17.3
20	Social tolerance ln Brain size Social tolerance*ln Brain size	4,14	0	3.96	+ - -	0.054 0.263 0.242	139.6	21.2
21	Social tolerance Ln Brain size	3,14	0	4.93	+ -	0.083 0.173	137.6	19.2
22	**Allomaternal care** ln (Number of immatures +1) Allomaternal care* ln (Number of immatures +1)	4,14	0	17.64	+ + +	**<0.001** 0.508 0.605	124.8	6.4
23	**Allomaternal care** ln (Number of immature offspring +1)	3,14	0	28.15	+ +	**<0.001** 0.529	121.2	2.8

Supplementary Table 5: Phylogenetic least squares (PGLS) models predicting proactive prosociality in all primates, i.e. including *Homo sapiens.*

number	**Model selection process:**	K,N	F	Direction	p-values	AICc	Delta AICc
1	ln Brain size	2,15	0.12	-	0.733	155.3	29.3
2	Cooperative hunting	2,15	0.015	+	0.906	155.4	29.4
3	Male bonds/Fission fusion	2,15	0.023	+	0.881	155.4	29.4
4	**Female bonds**	2,15	5.34	-	**0.038**	150.3	24.3
5	**Pair bonds**	2,15	12.73	+	**0.003**	145.2	19.2
6	**Social tolerance**	2,15	10.15	+	**0.007**	146.8	20.8
7	**Allomaternal care**	2,15	79.82	+	**<0.001**	126.0	0
8	**Allomaternal care** Social tolerance Allomaternal care*Social tolerance	4,15	27.32	+ + +	**0.004** 0.202 0.243	130.5	4.5
9	**Allomaternal care** Social tolerance	3,15	38.53	+ +	**<0.001** 0.505	128.6	2.6
10	**Allomaternal care** Pair bonds Allomaternal care*Pair bonds	4,15	25.7	+ + +	**0.023** 0.299 0.309	131.3	5.3
11	**Allomaternal care** Pair bonds	3,15	37.55	+ +	**0.0001** 0.663	128.9	2.9
12	Pair bonds Social tolerance Pair bonds*Social tolerance	4,15	4.00	+ + +	0.205 0.447 0.888	151.4	25.4
13	Pair bonds Social tolerance	3,15	6.53	+ +	0.175 0.431	147.6	21.6
14	**Allomaternal care** Female bonds Allomaternal care*Female bonds	4,15	25.72	+ - -	**0.003** 0.271 0.369	131.2	5.2
15	**Allomaternal care** Female bonds	3,15	38.53	+ -	**<0.001** 0.505	128.6	2.6
16	Pair bonds Female bonds Pair bonds*Female bonds	4,15	singular				
17	**Pair bonds** Femal bonds	3,15	5.88	+	**0.048** 0.955	148.4	22.4
18	Social tolerance Female bonds Social tolerance*Female bonds	4,15	3.45	+ - -	0.082 0.367 0.515	152.5	26.5
19	Social tolerance Female bonds	3,15	5.18	+ -	0.073 0.469	149.3	23.3

Supplementary Table 6: Comparison of the five different quantifications of allomaternal care in predicting proactive prosociality.

Data set	**Isler & van Schaik 2012, updated**	**Isler & van Schaik 2012**	**% allocare from Ross & MacLarnon 2000**	**Groups Ross & MacLarnon 2000**	**Broad categories**
All species, n=15	r=0.927, n=15, p<0.001	r=0.766, n=13, p=0.002	r=0.845, n=10, p=0.002	Rho=0.805, n=1, p=0.001	Rho=0.815, n=15, p<0.001
Without *Homo sapiens*, n=14	R=0.911, n=14, p<0.001	R=0.776, n=9, p=0.002	R=0.924, n=9, p<0.001	Rho=0.804, n=12, p=0.002	Rho=0.782, n=14, p=0.001

Supplementary Table 7: Overview of the performance per group regarding criteria 1-4

Group	N (w/o infs.)	Previous experimental experience?	% of transfers by individuals passing crit. 1	Number of tray pulling in test sessions 4 and 5 of phase IV	Number of tray pulling in control sessions 4 and 5 of phase IV	Criterion 2: progress of deliveries	Criterion 3: signs/signals necessary?	% of transfers by individuals passing crit.4
Lemur catta	8	yes	96.15%	30	13	decreasing	NA	NA
Varecia v.	5	yes	96.6%	13	5	decreasing	NA	NA
Ateles g. 1	13	yes	NA	0	0	low throughout	NA	NA
Ateles g. 2	10	no	64.93%	84	28	decreasing	no	NA
Pithecia p.	5	no	100%	92	9	sustained	no	100%
Saimiri s.	15	no	96.09%	110	18	decreasing	no	NA
Callithrix j. K	7	yes	100%	157	18	sustained	no	100%
Callithrix j. Jo	6	no	93.5%	129	27	sustained	no	74.89%
Callithrix j. V	7	yes	98.1%	84	17	sustained	no	98.1%
Callithrix j. Ju	6	yes	73%	36	6	decreasing	no	NA
Callithrix j. M	4	no	100%	76	21	sustained	no	81.11%
Saguinus o.	5	no	97.73%	128	10	sustained	no	71.4%
Leontopithecus	5	no	77.03%	145	67	sustained	no	89.73%
Cebus apella 1	5	yes	NA	0	0	low throughout	NA	NA
Cebus apella 2	10	yes	95.55%	107	82	decreasing	NA	NA
Cebus apella 3	8	no	NA	1	3	low throughout	NA	NA
Macaca f.	10	no	NA	54	42	low throughout	NA	NA
Macaca s.	6	no	NA	30	0	low throughout	NA	NA
Hylobates syn.	3	no	100%	92	19	sustained	no	100%
Hylobates lar	2	yes	100%	39	3	low throughout	NA	NA
Pan trogl.	29	no	65%	60	21	decreasing	no	NA

Supplementary Table 8: Correlations between pulling during test sessions 4&5 (*pulls test 4&5*), during control sessions (*pulls control 4&5*), the resulting *difference scores*, proactive prosociality (percentage of food provisioning in test session 4 and 5 of phase IV), both when including individuals who had not passed criterion 1, (*% transfers in test 4&5*) or excluding them *(% transfers in test 4&5, corrected*, i.e. the response variable used in the phylogenetic analyses) and the extent of *allomaternal care*, at species (n=14) and group level (n=21).

		% transfers in test 4&5	**% transfers in test 4&5, corrected**	**Allomaternal care**	**Difference score**	**Pulls control 4&5**
Pulls test 4&5	*Species: r*	**0.850**	**0.840**	**0.632**	**0.902**	**0.562**
	Species p	**<0.001**	**<0.001**	**0.002**	**<0.001**	**0.008**
	Group: r	**0.900**	**0.898**	**0.843**	**0.906**	0.476
	Group: p	**<0.001**	**<0.001**	**<0.001**	**<0.001**	0.085
Pulls control 4&5	*Species: r*	0.336	0.321	0.170	0.150	
	Species p	0.136	0.155	0.462	0.516	
	Group: r	0.381	0.373	0.359	0.060	
	Group: p	0.179	0.189	0.207	0.838	
Difference score	*Species: r*	**0.902**	**0.837**	**0.667**		
	Species p	**<0.001**	**<0.001**	**0.001**		
	Group: r	**0.839**	**0.840**	**0.784**		
	Group: p	**<0.001**	**<0.001**	**0.001**		
Allomaternal care	*Species: r*	**0.777**	**0.707**			
	Species p	**<0.001**	**<0.001**			
	Group: r	**0.925**	**0.927**			
	Group: p	**<0.001**	**<0.001**			
% transfers in test 4&5, corrected	*Species: r*	**0.999**				
	Species p	**<0.001**				
	Group: r	**0.999**				
	Group: p	**<0.001**				

Supplementary Discussion

Testing the dyad-specific predictions

It can be argued that dyad-specific patterns are blurred when tested in a group setting. For instance, the presence of numerous non-prosocial dyad types in a group might mask dyad-specific patterns of proactive prosociality if particularly intolerant individuals prevent the expression of these dyad-specific preferences. In practice, however, due to the extended period of testing, there were frequent opportunities for specific dyads to participate because in groups with low levels of transfers, dominant individuals soon lost interest in the apparatus, except during the motivation trials.

Likewise, the information value of the distribution of transfers within a group can be compromised when ceiling effects prevail, i.e. when transfers occurred in almost all trials, so that individuals who may have been motivated to provide food did not have the opportunity. This was the case in some callitrichid groups and human children. In two callitrichid groups with particularly high pulling rates (lion tamarins and Jojoba Group of the common marmosets), we therefore removed the individual responsible for the largest proportion of the transfers after the regular test sessions had ended and conducted additional trials. In both cases, other individuals would step up, resulting in similarly high provisioning rates (lion tamarins: 98.6% in the last regular test session compared to 90.01% in the additional test session when the top puller was removed from the group; common marmosets: 75.7% compared to 69.2%).

Ceiling effects could have similarly masked dyadic preferences in the human children, where all groups delivered food in almost all trials. Furthermore, we observed the emergence of self-organizing activities in the human groups, which resulted in higher-level cooperative patterns. In both groups of 5-year-olds, the children would start cueing at the donor position, deliver food, move on and become the recipient in the next trial, and finally move on to wait in line behind the donor position again (Supplementary Figure 6). Towards the end of their sessions, the 6-year-olds began to collect the delivered rewards in a plastic bag and store it to share it later with the other half of the class that would only be there in the afternoon, and was not taking part in the experiment.

How do the results relate to earlier studies on proactive prosociality in nonhuman primates?

Over the last decade, a large number of prosociality studies with nonhuman primates have been conducted (for recent reviews, see[47-49]). Whereas targeted helping tasks measure reactive prosociality[50] and are assumed to be cognitively demanding[51], dyadic prosocial games aim at assessing proactive prosociality and can be cognitively less demanding[52].

A general pattern emerging from these studies is that great apes, in particular chimpanzees, show prosocial behaviour in targeted helping tasks where individuals are given the opportunity to provide instrumental help to obtain a goal (but see[53], reporting a lack of targeted helping in chimpanzees and bonobos), but fail to show prosocial behaviour in tasks of proactive prosociality in which they can make decisions regarding the distribution of rewards, also called prosocial games (but see[54] for positive evidence in chimpanzees).

During human cognitive evolution, proactive, rather than reactive, prosociality arguably played a key role, because it may have paved the way towards shared intentionality and its cascading effects on cognition[55], resulting in the focus of the present study on proactive prosociality. In contrast to targeted-helping tasks, which have been used almost exclusively with great apes[47-49,51], dyadic prosocial games have also been applied to several monkey species. The pattern of results of proactive prosociality studies is inconsistent (see reviews above), and often contradictory because different studies may report different results for the same species (e.g. chimpanzees[54,56,57]; longtailed macaques[58,59]; cotton-top tamarins[60-62]) and sometimes even for the same individuals[61,62]. Some of the results are unexpected given the natural history of a species, such as the presence of proactive prosociality in highly despotic species[58], or the direction of prosocial behaviour from dominants to subordinates only[58,63].

Among the factors hypothesised to be responsible for the heterogeneity in the data are the social relationship between donor and recipient, and most importantly, the features of the task[48]. For instance, it could be shown that human children, at an age when they reportedly show proactive prosociality, fail a prosocial game commonly used with nonhuman primates due to cognitive task demands[52]. A major problem in the field is thus that it has not been possible so far to disentangle methodological artefacts from true species differences.

Even if we assume that all studies did validly measure proactive prosociality, it remains problematic to use this data to conduct comparative analyses aimed at testing functional hypotheses, for various reasons. First, because tests demonstrably[52] vary with regard to cognitive affordances, a positive bias favouring larger brained species ensues. Second, to conduct dyadic tests, one must separate individuals from the rest of the group. This is likely to affect group-living species in general, but different species and different individuals may be affected to different degrees. For instance, separation from the group is arguably most detrimental in highly interdependent species, in which individuals are highly motivated to rejoin their group rather than paying attention to the experimental situation. The same holds for dominant individuals, who normally control the interactions in their group. Finally, many studies aimed to produce a 'proof-of-principle'. Thus, to maximise the likelihood of finding positive evidence, dyads were tested in the direction most likely to show proactive prosociality[64], or dyads were preselected based on other criteria such as their propensity to pay careful attention to others' behaviour[65]. Whereas such studies are useful to detect the presence of proactive prosociality within a species, they obviously provide biassed estimates of a species' tendency towards it, making it virtually impossible to engage in meaningful comparisons with studies of other species where such preselection of subjects did not occur or was based on different criteria. (Below we will discuss why some selected dyads may show proactive prosociality in otherwise despotic species.)

The group service approach was developed and optimized for providing valid data for species comparisons[38]. First, we ensured that the cognitive requirements were minimal, by making sure there was only a small number of elements an individual would have to pay attention to simultaneously[52], and by providing evidence beforehand that all species, even the small-brained common marmosets, were able to understand the experimental setting as well as the consequences of their pulling[38]. Second, the animals were tested in their naturalistic group, without being separated, thus providing an ecologically more valid measure of the prevalence of proactive prosociality under naturalistic social conditions. Third, we provide

data representative for naturalistic groups, rather than preselected dyads that, in extreme cases, may not even share food passively[66] or tolerate each others' proximity under naturalistic situations[67]. On the one hand, this is relevant in order to provide data that is truly comparable between species. On the other hand, assessing proactive prosociality as it occurs under naturalistic situations is the most important measure for the test of functional hypotheses because ultimately, behavioural predispositions can only be favoured by evolution if they are actually expressed, rather than dormant and only expressed in artificial testing situations.

The optimization of the group service task to provide data suitable for broad species comparisons inevitably comes with the cost of decreased power to reliably quantify within-group, individual-level variation in proactive prosociality. For a complete picture, where we also obtain full information on within-species variation in proactive prosociality, a combination of group service tests with systematic testing of all dyads in all groups would be necessary.

But how do results from the group service task relate to earlier findings? As argued above, it is difficult to disentangle how different dyadic prosocial games relate to each other, and any conclusion that goes beyond speculation will require additional empirical data. The same obviously applies too to the relationship between group service and other prosocial games. Furthermore, we don't claim that group service captures all kinds of prosocial behaviour, but presumably refers to prosocial low-cost donations of food. Nevertheless, some insights from the present set of results may be possible, in combination with recent findings that inhaled oxytocin causally promotes social donation behaviour in despotic rhesus macaques[68]. Our results show that proactive prosociality varies on a continuum, rather than being uniquely present in strictly obligate cooperative breeders and being completely absent in non-obligate cooperative breeders. This is consistent with the idea that some basic predisposition towards proactive prosociality is present in nonhuman primates in general, perhaps emanating from the mother-infant context, with its expression regulated by the oxytocin system, including humans[69] and callitrichid monkeys[70], and macaques[68]. This scenario would be consistent with the occasional documentation of proactive prosociality in prosocial games in independently breeding primates, in particular when dyads preselect for dyad quality are involved.

If a general hormonal mechanism is responsible for increased levels of prosociality in cooperatively breeding primates, it is easy to understand the apparent lack of dyad-specificity in proactive prosociality in cooperatively breeding primates. Evolutionarily, natural selection can readily increase oxytocin levels or oxytocin-sensitivity, leading to a general, largely dyad-independent increase in proactive prosociality. This regulatory mechanism can thus explain why even though kin selection plays an important role in the establishment of cooperative breeding in a given species, the behavioural mechanism, i.e. proactive prosociality, no longer shows biased deployment towards kin but extends to all in-group members. In the species involved, groups mostly contain kin, but where non-kin are present, they are equally prosocial[71], so it is apparently not maladaptive to behave prosocially in an indiscriminate way towards all in-group members regardless of whether they are kin or not. In contrast, in non-interdependent groups of independent breeders, a generalized proactive prosociality towards any in-group member must have detrimental effects on individual fitness and therefore is

unlikely to be stable[49]. In this case, additional, perhaps cognitive, control mechanisms are needed to ascertain dyad-specificity of prosocial acts.

This scenario is also consistent with the specific case of humans. As in other cooperatively breeding species, the default-state in humans must be expected to be a prosocial attitude towards in-group members, as we find it in young children. In older children, cognitive and cultural influences become active that restrain prosocial behaviour in non-adequate conditions[4].

Supplementary Methods

Apparatus and procedure for human children

Human subjects were included in our sample to validate the group service paradigm. Specifically, we assessed whether this paradigm can empirically replicate human proactive prosociality, which has been shown to be exceptional in many different experimental paradigms[1]. We tested human children rather than adults (i) because adults are arguably confused when tested in simple, non-verbal paradigms designed for nonhuman primates and tend to perform in non-characteristic ways[2,3] and (ii) because cultural influences that can shape prosocial behaviours are expected to be more pronounced in adults than in children. The children included in our study were between 4:5 and 7:1 years old. We thus tested human children at an age when cultural influences most likely had already started to exert a measurable influence on prosociality[4]. However, proactive prosociality has been repeatedly demonstrated in children much younger than in our study[5,6], and the first cultural influences that are tractable around the age of 6-7 years tend to decrease, rather than increase, costly prosociality[4]. Younger children could not be tested in the present group setting, because this would have required the intervention of adults, which we wanted to avoid.

When testing the human children, we aimed at minimizing socially desired responses by (i) testing them in the absence of teachers and other authority figures and (ii) using an automated version of the group service apparatus (Supplementary Figure 1). The basic setup remained the same, but instead of a human experimenter providing the food on the apparatus, food could be dropped using a remote control, and instead of verbally announcing the dropping food, a signal light and an artificial sound would announce the next trial. The board with the food cups (blue board in Supplementary Figure 1) was inside a Perspex box, with two holes. The holes were placed in a way that when the apparatus was pulled, the food bowls in position 0 and in position 1 could be reached. A handle on the left hand side poking through the Perspex cover allowed the children to pull the blue board within reach. An experimenter sitting in the test room was remotely controlling the apparatus and taking notes. She would respond to the children if they addressed her with a general question but would avoid saying anything about the apparatus. When the children asked about the apparatus or what to do with it, she would respond "Yes, this is an apparatus for the children, and you're allowed to use it".

For the human children, it was not possible to run all control sessions of phase IV as well as phase V. As soon as they realized that on a given day, rewards were available only every 5 minutes (i.e. during motivation trials), they lost interest in the apparatus, leaving them

without meaningful occupation in the small room. It was not possible to keep the groups of children restricted in the test room for 5 (phase IV) and 10 (phase V) sessions of 70 min each in the absence of a teacher and without interfering by the experimenter. Instead of running all control conditions, we therefore tested the children's understanding of the apparatus in individual tests, conducted after phase IV. The aim of these individual tests was to assess whether the children understood that they could not reach the food themselves, and to exclude that they had a non-social preference for pulling the food in the distant position 1, e.g. because they thought the challenge was to figure out how to make food in position 1 available.

In the first part of the individual test, the children were presented with 2 test and 2 motivation trials as in phase IV, and they passed this part if they managed to retrieve the food in the motivation but not the control trial. In part two, the subjects were exposed to 10 test and 10 motivation trials and given the option to decide when the next trial should start, so that they could abort test trials until the next motivation trial was due. In the final part, they were allowed to choose where the reward would drop in the next trial, i.e. whether a test or a motivation trial would follow. We measured whether they were able to retrieve the food in test but not in motivation trials, whether they would abort test trials in order to more quickly move to the next motivation trial, whether they mentioned that the reward was not attainable in test trials, whether they mentioned that another child would be necessary to take the food in test trials, and whether they had a preference for motivation over test trials.

In the first part, all children managed to get food in the motivation trial, but two of the seven year olds managed to manipulate the apparatus in a way to also retrieve the reward in the test trial. All children also met one or more of the other criteria, except one of the seven year olds who had managed to retrieve the food in the more distant position. Accordingly, this subject was considered as not passing phase V.

Variables included in the regression models

As measure of proactive prosociality, the response variable in our models, we used the corrected percentage of food provisioning in the last two test sessions in phase IV for each species. The last two test sessions were chosen because by then, all subjects had had ample opportunity to learn that they could not get the food by themselves by pulling the apparatus, which in the very first trials may have led to false positives. The last two test sessions thus yield the most conservative estimate of proactive prosociality, but the results do not change if we take the total percentage of food provisioning over all test sessions. Furthermore, the percentage was corrected by including in this measure only provisioning by individuals who passed the criterion of pulling significantly more in test vs. control sessions. Thus, we used the percentage of food provisioning in test session four and five by those individuals who pulled more in test vs. control sessions. To calculate this percentage, trials were excluded in which food was provisioned to group members by individuals who had not passed this criterion. Including or excluding individuals who did not pass the above criterion did not change the result in the main analyses because both measures are strongly correlated (see also Supplementary Figure 2). For species where we had tested multiple groups, we used the average of the corrected percentages in sessions four and five to calculate one single value for each species. Aggregating data at the species level was necessary in order to be able to control for phylogenetic non-independence with PGLS. In sum, proactive prosociality measured with

the group service paradigm in the current study always refers to the most conservative measure, i.e. the corrected percentage of food provisioning in the last two test sessions per species.

As a proxy for cognitive ability, we included overall brain size in the analyses, because it best reflects *"g"* as a proxy of general intelligence[7,8]. "*g*" refers to a combined measure of cognitive performance in nonhuman primate taxa, derived from a large set of cognitive tests across nonhuman primates. Brain size data was taken from[9]. Brain size measures that control for body size where not used in the final analyses. However, they don't change the pattern of results if used instead of whole brain size data.

Social tolerance was assessed in phase II of the group service test, as described in the methods section. The presence of fission-fusion social organization, as well as cooperative hunting, strong male and female bonding and pair bonds was taken from compendia of primate behaviour[10,11]. These coded variables are presented in Supplementary Table 1. Note that in our sample, the presence across species of strong male bonds and fission fusion organization coincide.

The extent of allomaternal care has been quantified by Ross and MacLarnon[12] and more recently by Isler and van Schaik[9]. Ross and McLarnon's classification is based on the extent of infant carrying by non-mothers, whereas Isler and van Schaik's factor PC1 is deduced from a principal components analysis (PCA) of 93 primate species based on the frequency of the allomaternal behaviours allonursing, babysitting, provisioning by males, provisioning by others, carrying by males, and carrying by others. We calculated an updated version of this score that also included (i) species tested with the group service test but absent in the Ross and MacLarnon[12] or Isler and van Schaik[9] data set, and (ii) data from additional sources that were not available or not used in previous studies. The updated PC1 factor scores were calculated based on the PC1 factor loadings from the PCA over all 93 species by Isler and van Schaik, according to the following formula, where the extent of the various allomaternal behaviours can vary between 0 and 1, according to the definitions provided in[9]:

PC1 = 0.034*allonursing+0.729*babysit+0.613*protect+0.938*male provision+0.891*others provision+0.828* male carry+0.749*others carry.

In order to test the stability of the results, we classified the species into three broad categories, i.e. strictly independent breeders (category 1), strictly cooperative breeders (category 3), and species with intermediate breeding systems (category 2). These measures are compiled in Supplementary Table 2. All five measures of allomaternal care are strongly correlated (Supplementary Table 3). Subsequent analyses were based on the updated Isler and van Schaik[9] measure because it provides continuous data and has measures for all species tested with the group service paradigm.

Finally, we also included group size and the number of immature offspring (i.e. infants and juveniles) as potentially confounding variables in our models, where we used averages for those species where multiple groups had been tested.

Model selection for phylogenetic least squares (PGLS) models

To compare morphological or behavioural traits between species, their phylogenetic relatedness has to be considered, as similarity in traits is likely to result from shared ancestry. Phylogenetic least-squares analysis (PGLS) is commonly applied to account for phylogenetic non-independence[33]. In PGLS, the amount of phylogenetic structure in the model is estimated with a maximum likelihood calculation of the parameter lambda, which can take values between 0 (no phylogenetic structure, model equivalent to non-phylogenetic linear model) and 1 (strong phylogenetic structure, model equivalent to classic independent contrasts[34]). We used the caper package[35] in R[36] to fit PGLS models. Phylogenetic relationships and molecular branch length estimations were taken from the primate species-level supertree of Perelman et al.[37]. In the present data set, lambda was estimated to be zero in most models, indicating a low degree of phylogenetic structure in the data. The results are therefore mostly identical to non-phylogenetic least-squares regression models.

Model selection was performed following an information theory approach. The model with the lowest AICc (Akaike's Information Criterion, adjusted for small sample size) was chosen as the best model. However, due to the relatively small sample size we only considered models with one or two factors and their interaction effects. Two-factor models were considered for those factor pairs that independently had a significant effect on proactive prosociality in unifactorial tests. All variables have been centred before being analysed, and brain size was logarithmized. Note that the AICc reported in caper is slightly different from the AICc of other packages such as glmulti or AICcmodavg, as the degree of freedom of the fitted models is one more in the latter. This difference does, however, not affect the model selection results.

PGLS models excluding *Homo sapiens:* Supplementary Table 4 shows the model selection process for phylogenetic least squares (PGLS) models predicting proactive prosociality in nonhuman primates (N=14 species, excluding *Homo sapiens*). Among all models, the unifactorial model with allomaternal care as predictor (number 7) explains most of the variation in proactive prosociality..

PGLS models including *Homo sapiens:* Supplementary Table 5 shows the model selection process for phylogenetic least squares (PGLS) models predicting proactive prosociality in all tested primates (N = 15 species, including *Homo sapiens*). Lambda was estimated to be 0 in all models. The best model was, as above, the one that included only the extent of allomaternal care.

Validity: Do we really measure proactive prosociality with the group service experiment?

To demonstrate that food deliveries in the context of group service are due to proactive prosociality, four criteria have to be fulfilled[38]. First, individuals responsible for the transfers must understand the apparatus, and not pull the board more often in control than in test sessions of phase IV. Second, deliveries have to persist over the 5 test sessions, rather than decline; declining levels of deliveries over time would indicate that deliveries in the beginning are accidental and gradually discontinued as the animals learn they do not acquire any food this way. Third, donors must not be prompted by the potential recipients to pull the board, e.g. by signalling need, begging, or harassment. Finally, subjects have to pass phase V, providing additional proof that pulling did not result from a lack of understanding that they themselves

could not reach the food. These criteria are conservative, i.e. applying them rigorously decreases the risk of false positives but increases the risk of wrongly excluding true positives.

Criterion 1: Pulling rates in test vs. control sessions – According to the first criterion, subjects have to pull significantly more frequently in test than in control sessions (based on individual Chi^2 tests). We therefore coded for each individual in phase IV whether she met this criterion and only used transfers of those individuals who passed this criterion to estimate proactive prosociality. Comparing the percentage of all transfers with the percentage of transfers by individuals fulfilling criterion 1 reveals that both measures are very similar and highly correlated in all test sessions (least significant result in test session 2: $r = 0.996$, $n = 15$, $p<0.001$; Supplementary Figure 2 shows the correlation of both measures for test session 4 and 5 which were used for the main analyses) and both perform similarly when used as response variable in the regression models. Subjects not passing this criterion were mostly subordinates who only had access to the apparatus during control trials. Supplementary Table 7 (column 4) provides the percentage of transfers that were performed by individuals passing criterion 1.

In dyadic prosocial games, proactive prosociality is typically expressed as difference score, i.e. the frequency of pulling the board in the test condition minus the frequency of pulling it in the control condition. Individual difference scores can thus be calculated in the group service test, where in test conditions, pulling the tray can lead to a reward for one of the other group members, but in control conditions, it does not. We thus calculated difference scores to assess whether the main results remain stable. In order to allow comparability of the data with the main analyses, we calculated the difference scores from test sessions 4 and 5 and control sessions 4 and 5 of phase IV, by calculating the difference between the pulling frequencies in test sessions (supplementary Table 7, column 5) minus the pulling frequencies in control sessions (supplementary Table 7, column 6), at the group and species level. Note that when calculating the difference scores for all 5 test sessions, the pattern of results remains the same.

Supplementary Table 8 shows that the difference score is strongly correlated with the percentage of food provisioning in test session 4 and 5 of phase IV, both when excluding individuals who had not passed criterion 1 (i.e. the response variable used in the phylogenetic analyses) or including them. Furthermore, the difference score was again positively correlated with the amount of allomaternal care. Importantly, these positive correlations arose because pulling during test sessions, but not during control sessions, was correlated with the percentage of food transfers and cooperative breeding. Thus, cooperatively breeding species were not simply more likely to pull the board overall, but did so only in the test sessions.

The advantage of differences scores is that they control for baseline rates of pulling. Nevertheless, difference scores are not fully appropriate in the group service context for at least two reasons. First, the duration of trials in test vs. control conditions is not the same. Test trials are aborted as soon as food is taken, or after 1 min, whereas control trials always last for one minute. A direct comparison thus produces a conservative bias, and individuals who pull more during test sessions do so despite the fact that test sessions may have been considerably shorter. However, because the duration of test trials was different for the different groups (depending on how often they did provide food), the difference scores reflect different

measures in different groups, because the above mentioned conservative bias did not apply to the same extent in all groups (i.e. in groups where no provisioning occurred, all test trials lasted 1 min and all control trials lasted 1 min, but in cotton top tamarins, for instance, a test trial lasted on average only 12.24 sec ±10.8). Thus, difference scores cannot be properly compared between groups or species.

Second, mere pulling in experimental conditions may lead to a food transfer, but if the puller is not coordinating its pulling with a potential recipient, it will not. The number of actual transfers in the test sessions is thus at best the same, but usually less than the number of pulls in the test sessions because the board can roll back if released before a partner takes the food. Thus, even though highly correlated (Suppl. Table 8), the number of food transfers is a better proxy for prosociality than the number of pulls in the experimental sessions.

Criterion 2: Sustained vs. decreasing rates of provisioning – Species can be classified according to the progress of deliveries over the five test sessions. Sustained low levels require no further interpretation. Declining levels over the five test sessions most likely refer to false positives in the beginning, whereas sustained provisioning represents the strongest evidence for proactive prosociality.

Three species showed deliveries that were low throughout the test (less than 20% of deliveries in each of the five sessions): gibbons, Japanese macaques, lion-tailed macaques. Six species showed variable rates in the beginning but decrease towards the end of the test phase (i.e. had negative correlation coefficients of proactive prosociality over the 5 test sessions: chimpanzees [Rho = -0.9], spider monkeys [Rho = -0.4], capuchin monkeys [Rho = -0.7], squirrel monkeys [Rho = -1.0], ringtailed lemurs [Rho = -0.4], ruffed lemurs [Rho = -0.78]). Finally, six species showed intermediate (> 30 %) to high levels of deliveries right from the beginning that were sustained over all 5 test sessions (i.e. had non-negative correlation coefficients: common marmosets [Rho = 0.9], cotton-top tamarins [Rho = 0.56], lion tamarins [Rho = 0.8], siamangs [Rho = 0.2], sakis [Rho = 0.5] and human children [Rho = 0.3]. The progress of deliveries per group is provided in Supplementary Table 7 (column 7).

These three groups identified based on the development of transfers over time differ in the expected direction with regard to the extent of allomaternal care (one-way ANOVA $F(12,2)=37.134$, $p<0.001$), but not social tolerance ($F(2,12)=2.88$, ns; Supplementary Figure 3). Thus, this analysis confirms the main conclusion of the study: that species with the strongest evidence for proactive prosociality according to the second criterion also have the highest amount of allomaternal care.

Criterion 3: Signs and signals of need are not necessary – Whenever a transfer is initiated by the donor, without being prompted by signs or signals from a potential recipient, we can be sure that such signs and signals of need (e.g. reaching for the food, begging, harassment) are not necessary for the initiation of the transfer, which was therefore truly proactive, thus satisfying criterion 3.

We quantified the influence of signs and signals of need from potential recipients both within and between species. For each trial in test session 5 of phase IV, we coded whether signs or signals of need occurred and whether food was provisioned or not. Signs/signals were only coded as present if they could be perceived by a potential donor. The most obvious sign

of need by recipients are reaching attempts, i.e. extending one's arm (or tail, in the case of spider monkeys) out of the cage towards the food in position 1. Behaviours directly signalling need, such as begging towards potential pullers or attempts to coerce another individual to pull the handle were never observed in any species. Our analyses where therefore limited to the influence of reaching attempts.

Supplementary Figure 4 shows the percentage of transfers in trials where reaching occurred compared to trials in which no reaching was observed, for those species where there were enough transfers for analysis, i.e. rates higher than 10% (for species with multiple groups, we present averages). In all these species a portion of transfers occurred in the absence of signs and signals of need, thus clearly classifying as proactively prosocial. Supplementary Table 7 (column 8) summarizes whether signs or signals of need where necessary to release food transfers for each group.

Particularly conservative evidence for proactive prosociality is present if reaching does not increase the likelihood of transfers. Reaching did not increase the likelihood of provisioning in most of the species where we could test for this (chimpanzees, cotton-top tamarins, sakis, spider monkeys, common marmosets: Chi^2 always < 1.07, ns), but did so in gibbons ($Chi^2 = 17.13$, $p<0.001$). In several species, we could not test for reaching effects due to ceiling effects in transfers (in lion tamarins and siamangs) or because reaching was too common (in squirrel monkeys). Thus, with the exception of the independently breeding gibbons, expressing interest in the food did not affect food transfers, supporting the claim that we measure proactive, rather than reactive prosociality.

Since in various species, we were not able to assess the effect of reaching due to ceiling effects, we also repeated the between-species comparison for test session 5 of phase IV and included the percentage of trials in which reaching occurred as predictor variable. Reaching had no unifactorial effect on proactive prosociality (lambda= 0, $F(1,13) = 0.54$, n.s, AICc from PGLS = 45.9). Allomaternal care again was the strongest predictor for proactive prosociality in session 5 (lambda = 0.691, $F(1,13) =76.78$, $p < 0.001$, AICc from PGLS = 19.8). Thus, reaching was not necessary to elicit food transfers, and the amount of reaching did not explain variation in transfers between species, just as it did not within species.

Criterion 4: Performance in phase V – For those species who passed criteria 1-3 and continued to provide food to group members in more than 40% of all cases in session 5 (common marmosets, lion tamarins, cotton-top tamarins, siamangs and sakis), we conducted an additional control phase V to test whether the subjects really understood the consequence of their pulling. Phase V was identical to phase IV, with the exception that a fine-meshed grid in front of the recipient bowl in position 1 prevented group members to take the food even if the board was pulled by the donor. Thus, if the board was pulled for any reason other than provisioning, for instance because the stimulus of the food in position 1 was simply too salient and the animals were therefore unable to inhibit pulling the board when seeing this reward, we would expect them to continue pulling at the same rate as in phase IV.

For each individual, we compared the pulling rates in the test sessions of phase V with the test sessions of phase IV. Criterion 4 was considered passed when the individual pulled significantly less in test sessions of phase V than phase IV. In the siamangs and the sakis, 100 % of the transfers in phase IV had been performed by individuals who subsequently passed

phase V, in common marmosets 86.57 %, in lion tamarins 89.73 % and in cotton-top tamarins 71.4 %. In the cotton-top tamarins, an individual who did not pass phase V had also started to pull more in control sessions of phase V compared to control sessions in phase IV, i.e. when no food was on the board at all ($Chi^2 = 22$, df = 1, $p < 0.001$). Thus, for some unknown reason, this individual had developed a habit of pulling the board during phase V. Thus, apart from the cotton-top tamarins where a single individual developed a preference for pulling the board in phase V, at least 85% of all transfers were performed by individuals who passed criterion 4 and understood the consequences of their actions, suggesting that they intentionally provided others with food.

Note that comparing pulling frequencies in test sessions of phase IV with phase V is highly conservative, since in phase IV trials ended whenever a transfer occurred, whereas in phase V, each trial lasted for 60 sec. During phase V, individuals thus had much more time to show the behaviour of pulling. This applies in particular to callitrichid monkeys, where transfers occurred at high levels and with short latencies during phase IV (e.g. mean trial duration in cotton-top tamarins, session 4 and 5: 12.24 sec ±10.8). Trials in phase IV were thus much shorter than in phase V and could last only a few seconds, while each trial in phase V lasted for 1 min, giving the subjects considerably more time to pull.

Expectation of reciprocation and reputation seeking – Finally, food provisioning in the group service context may be motivated by the expectation of reciprocation or reputation seeking. Whether it is ever possible to exclude expectation of reciprocation effects is intensely debated in humans[39,40], because such expectations may guide the behaviour implicitly even if no potential reciprocators are present in the experimental situation, or may instead be directed at the experimenters. In nonhuman primates, expectation of reciprocation effects are thought to be unlikely because calculated reciprocity is cognitively demanding and approaches of explicitly offering the possibility to establish reciprocation patterns between nonhuman primates typically fail, even in chimpanzees[41-46]. Even so, despite the difficulty of completely excluding the possibility of expectation of reciprocation, some of our results suggest that this mechanism was not at work: First, all groups in which transfers were low or absent altogether obviously did not make use of this mechanism. Second, over time, the establishment of reciprocation patterns would lead to a continuous increase in food transfers in phase IV, rather than a decrease. Thus, it is unlikely that expectation of reciprocation was involved in those groups with decreasing rates of provisioning over time. Third, since such effects are arguably cognitively demanding, we would expect a positive correlation between food transfers and brain size, or any other measure of cognitive ability, which we do not find. Fourth, removing the individuals with the highest rates of food provisioning in a group should have a disruptive effect on reciprocation patterns. However, the two experimental removals of the top pullers in lion tamarins and one group of common marmosets did not have such an effect (see below). Reputation seeking, finally, can be viewed as a generalization of calculated reciprocity from the dyadic context to the group context. To date, there is no evidence for a concern about reputation in any nonhuman primate in food sharing contexts, perhaps because cognitively, it may be even more demanding than expecting reciprocation by a single individual.

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